Aiden Tran

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Education

University of California, Berkeley — Berkeley, CA
BA, Computer Science — *Exp 2028*

- **Relevant Coursework:** Computer Programs, Data Structures, Computer Architecture, Algorithms, Discrete Math and Probability Theory, Linear Algebra and Differential Equations, Computational Astrophysics

Projects

ML Library & LLM Inference Engine | *C++17, AVX2 SIMD, Transformers* — GitHub

- Built a machine learning library from scratch in C++17, with no ML frameworks, implementing 11 model types (classical, neural networks, a transformer) and 6 optimizers to learn how ML works beneath PyTorch
- Predicted housing prices and heart disease risk from public Kaggle datasets using the library's regression models
- Ran a real 1B-parameter LLM (Llama 3.2) on the library's own transformer; GPU acceleration is 42x faster than CPU
- Cut training and inference time by up to 2.5x on large datasets by vectorizing matrix math with AVX2 SIMD

Multiplayer Fighting Game | *Lua, Rollback Netcode, ECS* — Technical Writeup

- Ran a two-week public test: 1,000+ players, 600K+ social media impressions; shipped updates from feedback tickets
- Eliminated online lag with rollback netcode, resimulating 12 frames in under 3 ms, 5x under the 60 fps budget
- Designed a deterministic game loop and entity-component system (60+ states per character) to keep players in sync
- Published a public technical article explaining the netcode design and challenges, to help other developers

Real-Time Audio Signal Processor | *C++, FFT, ImGui* — GitHub

- Built a C++ desktop app that identifies the musical note being played from a live microphone in real time, using a Fourier transform (Cooley-Tukey FFT) pipeline with framing, windowing, and peak detection
- Added an audio file mode that returns the note for each segment of a recording using the same pipeline
- Built a responsive ImGui interface with live waveform and frequency spectrum graphs showing what the app detects

Black Hole Spacetime Simulations | *Python, Pandas, Matplotlib* — GitHub

- Co-developed a Python physics simulation with a team of 6, modeling gravitational time dilation, redshift, and spacetime curvature around black holes using the Schwarzschild metric from general relativity
- Owned the time dilation and curvature modules, using black hole masses (AGN database) and Pandas
- Built interactive mass, radius, and observer controls plus 3D curvature and infalling-object visuals using Matplotlib

MarkovLuau | *Lua, Claude Code, DSL* — GitHub

- Led design and testing of an open-source port of MarkovJunior, built with Claude Code, to generate game levels
- Designed 3 features missing from the original: a code-based rule DSL, time-travel playback, and grid locking
- Validated compatibility by running the original MarkovJunior XML models unmodified through the included parser
- Published documentation with install guide, examples, and API reference (docs.aidentran.dev/markov-luau)

Experience

Key On Harmony | *Intern* — Jul 2024 – Aug 2024

- Built a tool in React Typescript that lets a client bulk upload data and flags formatting errors before entry
- Maintained the client's accounting records in Google Sheets as part of a consulting team

Skills

Languages: Python, C++, Lua, Java, C#, TypeScript, SQL
ML & Data: PyTorch, LangChain, NumPy, Pandas, Matplotlib, Transformers, LLM Inference
Web: React, Next.js, Node.js, Postgres
Tools & Engines: Git, Docker, Linux, CMake, Claude Code, Unity, Godot